HANDWRITTEN DIGIT RECOGNITION USING MNIST

PROJECT REPORT

ON

HANDWRITTEN DIGIT RECOGNITION USINGNEURALNETWORKS(MNISTDATASET)

Submitted in partial fulfilment of the requirementsfor

[ARTIFICIAL INTELLIGENCE]

Submitted by

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Under the guidance of

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Academic Year: [2026–2027]

CERTIFICATE

This is to certify that the project report entitled “Handwritten Digit Recognition Using Neural Networks (MNIST Dataset)” has been prepared by **Anushka Khaladkar ,**student of **ComputerEngineeringClass:TE** at **Padmabhushan Vasantdada Patil Institute of Technology, Bavdhan**, under my guidance.

The work presented in this report is submitted as part of the academic requirements for TE / ARTIFICIAL INTELLIGENCE. The project demonstrates the use of Python and a neural network to classify handwritten digit images from the MNIST dataset.

Project Guide: TEAM NAVIOTECH

Date: 30/10/2026

Place: PUNE

DECLARATION

I,Anushka Khaladkar, hereby declare that the project report entitled “Handwritten Digit Recognition Using Neural Networks (MNIST Dataset)” has been prepared by me for academic purposes.

I have made an effort to understand the dataset, preprocessing steps, neural network architecture, model training, evaluation, and prediction process. Any external material used for learning or reference is acknowledged in the References section.

The results section must be completed using the actual outputs obtained when running the project. No accuracy value or screenshot should be presented as an actual experiment unless it was produced by the submitted code.

Name: Anushka Khaladkar

Date: 30/10/2026

ACKNOWLEDGEMENT

Iwould like to express my sincere gratitude to my project guide, [TEAM NAVIOTECH], for the guidance and support provided during this project. Their suggestions helped me understand the process of developing a basic artificial intelligence application.

I am thankful to the Department of Computer Engineering and Padmabhushan Vasantdada Patil institute of technology, bavdhan for providing the opportunity to work on this project. I also acknowledge the developers of Python, TensorFlow, Keras, NumPy, and Matplotlib for creating tools that make machine learning accessible to students.

Finally, I thank my family and friends for their encouragement and support throughout the completion of this project.

ABSTRACT

Handwritten Digit Recognition is an application of Artificial Intelligence (AI) and Machine Learning (ML) in which a computer identifies a numerical digit from an image. Human handwriting differs in size, shape, slant, and writing style, so recognizing digits automatically is a useful image-classification task.

This project uses the MNIST (Modified National Institute of Standards and Technology) dataset, which contains 70,000 grayscale images of handwritten digits from 0 to 9. The dataset is divided into 60,000 training images and 10,000 testing images. Each image has a size of 28 × 28 pixels.

The project is implemented in Python using TensorFlow and Keras. The images are normalized by dividing pixel values by 255. A feed-forward neural network is created with a flattening layer, a dense hidden layer using ReLU activation, a dropout layer, and an output layer with ten values corresponding to digits 0–9. The model is trained on the training data and evaluated on unseen test images.

The system displays a sample digit, reports test accuracy and loss, plots training and validation accuracy, and predicts the digit represented by a selected test image. The actual performance values should be recorded after running the notebook. The project provides practical experience with image data, preprocessing, neural networks, model evaluation, and prediction.

Keywords: Artificial Intelligence, Machine Learning, MNIST, Handwritten Digit Recognition, Neural Network, Python, TensorFlow, Image Classification.

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1. INTRODUCTION

Artificial Intelligence is a field of computer science concerned with building systems that perform tasks normally associated with human intelligence. Examples include recognizing images, understanding language, making recommendations, and identifying patterns in data. Machine Learning is a branch of AI in which a system learns patterns from examples rather than relying only on manually written rules.

Image classification is a machine-learning task in which an image is assigned to one category from a predefined set. In handwritten digit recognition, the categories are the ten numerical digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. Handwritten digit recognition is challenging because two people may write the same digit differently. A digit may be narrow or wide, tilted, thick, thin, or connected in an unusual way. An effective model must learn features that help distinguish one digit from another despite these differences.

The MNIST dataset is widely used for learning and demonstrating image classification. Its small grayscale images make it suitable for a beginner project because it avoids the need to collect and label a large custom dataset.

In this project, a neural network is trained using the MNIST training set. The trained model is then evaluated on the test set and used to predict the digit in an example image. The project is intended for learning and demonstration rather than as a complete commercial handwriting recognition product.

2. PROBLEM STATEMENT

Manual reading of handwritten numbers can be time-consuming when large numbers of forms, cheques, postal codes, or records must be processed. Differences in handwriting make it difficult to recognize every digit using a fixed set of rules.

The problem addressed by this project is to develop a basic AI system that accepts a grayscale image of a handwritten digit and predicts which digit, from 0 to 9, it represents. The system should learn from labeled examples, test its performance on unseen examples, and display a prediction that can be compared with the correct label.

Problem definition

Given an input imageofsize 28 × 28 pixels, classify the image into one of ten classes: 0–9.

Expected output

- A trained neural network model.
- A test accuracy and test loss value calculated from the actual run.
- A graph showing training and validation accuracy.
- A predicted digit for a selected test image.
- A reproducible notebook containing the implementation.

3. OBJECTIVES AND SCOPE

3.1 Objectives

- Understand the basic concepts of artificial intelligence and neural networks.
- Load and explore the MNIST handwritten digit dataset.
- Prepare image data for model training.
- Build a neural network using TensorFlow and Keras.
- Train the model using labeled examples.
- Evaluate model performance on unseen test images.
- Visualize training accuracy and validation accuracy.
- Predict the digit shown in a test image.
- Document the implementation, results, limitations, and possible improvements.

3.2 Scope

Theprojectfocuses on classifying individual grayscale digit images from the MNIST dataset. It includes data loading, normalization, model construction, training, evaluation, visualization, and prediction. It does not include a full optical character recognition system for paragraphs, a mobile application, or a production deployment.

3.3 Learning outcomes

After completingthe project, a student should be able to explain the dataset, identify the purpose of the main code sections, describe the neural network layers, interpret accuracy and loss, and demonstrate a prediction. The student should also understand that a model can make incorrect predictions and that test performance does not guarantee perfect performance on real-world handwriting.

4. LITERATURE REVIEW / BACKGROUND STUDY

Image classification has developed from handcrafted image features and traditional classification algorithms toward deep learning methods that learn useful features directly from images. Neural networks are made of connected computational units called neurons. During training, the network adjusts its parameters to reduce the difference between predictions and known labels.

The MNIST dataset became a common benchmark for handwritten digit recognition because it provides a consistent collection of labeled examples. Researchers and students can compare methods on the same training and testing data. The dataset is also convenient for demonstrating the full machine-learning workflow.

4.1Neuralnetworks

A neural network typically contains an input stage, one or more hidden layers, and an output stage. Each layer transforms its input using learnable weights and biases. An activation function introduces non-linearity so the model can learn more complex patterns.

4.2Denseneuralnetwork

A dense layer connects each neuron to all outputs from the previous layer. In this project, the 28 × 28 image is flattened into 784 values, which are passed to a dense hidden layer. The final dense layer produces ten class scores.

4.3 ReLU and dropout

ReLU, or Rectified Linear Unit, returns zero for negative inputs and keeps positive inputs. It is commonly used in hidden layers. Dropout randomly disables a portion of units during training to help reduce overfitting. It is inactive during ordinary evaluation and prediction.

4.4Evaluation

Accuracy is the proportion of predictions that are correct. Loss is a numerical measure of prediction error used during training and evaluation. Training metrics describe performance on examples used for learning, while test metrics estimate performance on data held out from training.

5. REQUIREMENTS AND TOOLS

5.1 Hardware requirements

- Acomputer,laptop, or compatible mobile device with a modern browser.
- Internet connection for opening Google Collab and downloading the dataset.
- Sufficient memory and processing resources to run a small neural network.

5.2 Software requirements

Tool / Library	**Purpose**
Python	Programming language used to implement the project.
Google Colab	Online notebook environment for writing and executing Python code.
TensorFlow	Machine-learning framework used to build and train the model.
Keras	High-level API used to define neural network layers and training.
NumPy	Numerical operations and selecting the predicted class.
Matplotlib	Displaying digit images and plotting accuracy graphs.

5.3 Why Google Collab?

Google Collab runs notebooksin a hosted environment and generally includes many common Python libraries. It reducessetup work for beginners. The notebook can be saved to Google Drive or downloaded as an .ipynb file. Internet access is normally needed to download MNIST on the first run.

5.4 Setup procedure

1. Open https://colab.research.google.com/ and sign in.
2. Create a new notebook and rename it MNIST_Digit_Recognition.
3. Run the code cells in order, from importing libraries through saving the model.
4. If a cell fails, read the error message and correct it before continuing.
5. Save the notebook and download a copy for submission.

6. DATASET DESCRIPTION

The MNIST dataset contains grayscale images of handwritten digits and a label for each image. The label is an integer from 0 to 9. The dataset is divided into training data and test data so the model can be evaluated on examples it did not use for learning.

Property	Description
Dataset name	MNIST Handwritten Digits Dataset
Total images	70,000
Training images	60,000
Test images	10,000
Image dimensions	28 × 28 pixels
Image type	Grayscale
Number of classes	10
Labels	Digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

6.1Inputrepresentation

Each image is represented as a two-dimensional array containing 28 rows and 28 columns. Each pixel has an intensity value. In the original dataset, pixel values range from 0 to 255. A value near 0 represents black and a value near 255 represents white, with intermediate values representing shades of gray.

6.2 Training and testing split

The model learns from the 60,000 training images and their labels. The 10,000 test images are kept separate for final evaluation. Using separate test data gives a more meaningful estimate of performance on unseen examples than evaluating only on the training images.

6.3Datasetlimitations

MNIST contains small, centered, grayscale digit images. Real-world images may contain colored backgrounds, shadows, varied image sizes, multiple digits, noise, and different writing tools. Therefore, performance on MNIST may not directly represent performance on photographs or scanned forms.

7. METHODOLOGY

The project follows a standard supervised-learning workflow. In supervised learning, each training input is paired with a correct label. The model makes predictions, compares them with labels through a loss function, and updates its parameters during training.

7.1Workflowstages

6. Data collection: load MNIST using the TensorFlow Keras dataset utility.
7. Data exploration: display a sample image and its correct label.
8. Preprocessing: convert image arrays to floating-point values and normalize pixel intensities.
9. Model design: define a sequential neural network with flattening, dense, dropout, and output layers. 10. Compilation: select the Adam optimizer, sparse categorical cross-entropy loss, and accuracy metric.
11. Training: fit the model using training images and labels for five epochs, using a validation split.
12. Evaluation: calculate loss and accuracy on the separate test set.
13. Prediction: pass one test image through the trained model and select the class with the largest score.
14. Visualization and saving: plot training and validation accuracy and save the model.

7.2Datanormalization

The original pixel values range from 0 to 255. Dividing each value by 255.0 converts it to the range 0.0 to 1.0. Normalization provides a consistent numerical scale and often helps neural networks train more effectively.

7.3Supervisedlearning

The dataset provides the correct digit label for each training image. The model learns the relationship between pixel patterns and digit labels. During evaluation, the predicted labels can be compared with the known test labels to calculate performance.

8. SYSTEM DESIGN

The system can be divided into five main modules: dataset loading, preprocessing, model training, evaluation, and prediction. Each module has a clear purpose and passes data to the next stage.

Module	Input	Process	Output
Datasetloading	MNISTsource	Load image arrays and labels	Trainingand test sets
Preprocessing	Pixel arrays	Convert to float and normalize	Normalized arrays
Model training	Images and labels	Learn parameters over epochs	Trained neural network
Evaluation	Test images and labels	Calculate loss and accuracy	Evaluation metrics
Prediction	One test image	Calculate class scores and choose highest	Predicted digit

8.1 System flow

Start → Load MNIST → Explore Image → Normalize Data → Build Model

↓

Compile Model → Train Model → Evaluate Test Set → Predict Digit

↓

Display Results and Save Model → End

This flow represents the order in which the notebook should be executed. If the notebook runtime is restarted, earlier cells may need to be run again before later cells can use the variables.

8.2 Inputs and outputs

The input to the classifier is a 28 × 28 normalized image. The output layer produces ten scores, one for each digit class. The program uses argmax to identify the index of the largest score. That index is interpreted as the predicted digit.

9. NEURAL NETWORK ARCHITECTURE

The model uses the Keras Sequential API, which stacks layers in a linear order. It is a simple feed-forward network appropriate for learning the basic concepts of image classification.

Layer	Configuration	Purpose
Flatten	Input shape 28×28	Changesthe two-dimensional imageintoa vector of 784 values.
Dense	128 units, ReLU	Learnscombinations of pixel patternsuseful for classification.
Dropout	Rate 0.2	Randomly drops a portion of units during training to help reduce overfitting.
Dense output	10 units, linear scores	Produces one score for each possible digit class.

9.1 Output scores

The output layer has ten units because there are ten digit categories. The model is compiled with from_logits=True, which means the output values are treated as raw class scores (logits). The sparse categorical cross-entropy loss function handles these scores appropriately during training.

9.2Optimizerandlossfunction

Adam is an optimizer that updates model parameters based on gradients and adaptive estimates of those gradients. Sparse categorical cross-entropy is suitable when the correct class is represented by an integer label such as 0, 1, or 7. Accuracy reports the fraction of examples classified correctly.

9.3Whystartwithadensenetwork?

A dense network is relatively easy to understand and implement. It provides a useful baseline for learning the training workflow. Convolutional neural networks often perform better on image tasks because they learn local spatial features, but they can be added as a future improvement.

10. ALGORITHM

10.1 Step-by-step algorithm

15. Startthe program.
16. Import TensorFlow, NumPy, and Matplotlib.
17. Load the MNIST training and test sets.
18. Display one training image and its label.
19. Convert the training and test image arrays to floating-point values.
20. Divide all pixel values by 255.0 to normalize them.
21. Create a sequential neural network with a Flatten layer, a 128-unit ReLU layer, a Dropout layer, and a 10unit output layer.
22. Compile the model with the Adam optimizer, sparse categorical cross-entropy loss, and accuracy metric.
23. Train the model for five epochs with 10% of the training data used for validation.
24. Evaluate the model on the test set.
25. Plot training and validation accuracy.
26. Select a test image and use the model to predict its digit.
27. Compare the predicted digit with the actual label.
28. Save the trained model.
29. Stop the program.

10.2 Pseudocode

BEGIN
Import required libraries
Load MNIST dataset
Normalize training and test images
Define neural network model
Compile model
Train model using training data
Evaluate model using test data
Display accuracy and graph
Predict digit for selected image
Save trained model

END

11. IMPLEMENTATION

The implementation is performed in Google Collab using Python. Run each code cell in the order given below. TensorFlow's Keras API downloads the MNIST dataset when the dataset-loading function is first called and the data is not already cached.

11.1 Import libraries

```
import tensorflow as tf import numpy as np import
matplotlib.pyplot as plt print("Libraries imported successfully!")
```

11.2 Load dataset

```
(x_train, y_train),(x_test, y_test) = (
tf.keras.datasets.mnist.load_data() ) print("Training
images:", x_train.shape) print("Testing
images:", x_test.shape)
```

11.3 View a sample

```
plt.imshow(x_train[0], cmap="gray") plt.title(f"Actual Digit:
{y_train[0]}") plt.axis("off") plt.show()
```

11.4 Normalize images

```
x_train = x_train.astype("float32") / 255.0 x_test =
x_test.astype("float32") / 255.0 print("Data preprocessing
completed!")
```

11. IMPLEMENTATION (CONTINUED)

11.5 Build and compile the model

```
model          =          tf.keras.Sequential([
tf.keras.layers.Flatten(input_shape=(28,   28)),
tf.keras.layers.Dense(128, activation="relu"),
tf.keras.layers.Dropout(0.2),     tf.keras.layers.Dense(10) ])

model.compile(     optimizer="adam",
loss=tf.keras.losses.SparseCategoricalCrossentropy(
from_logits=True     ),     metrics=["accuracy"] ) model.summary()
```

11.6 Train the model

```
history = model.fit(     x_train,
y_train,     epochs=5,     validation_split=0.1
)
```

11.7 Evaluate the model

```
test_loss, test_accuracy
= model.evaluate(     x_test,
```

```
y_test, verbose=0 )print("Test Accuracy:",  round(test_accuracy * 100,2),"%")

print("Test Loss:",

round(test_loss, 4))
```

11.8 Plot accuracy

```
plt.plot(history.history["accuracy"], label="Training Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy") plt.title("Model
Accuracy") plt.xlabel("Epoch") plt.ylabel("Accuracy") plt.legend() plt.grid(True)
plt.show()
```

11. IMPLEMENTATION (CONTINUED)

11.9 Predict a digit

```
index =
0 image = x_test[index]

prediction = model.predict(     np.expand_dims(image,

axis=0),     verbose=0 ) predicted_digit =
np.argmax(prediction[0])


plt.imshow(image, cmap="gray") plt.title(f"Predicted Digit:

{predicted_digit}") plt.axis("off") plt.show() print("Actual

Digit:", y_test[index]) print("Predicted Digit:",

predicted_digit)
```

11.10 Save model

```
model.save("mnist_digit_model.keras") print("Model saved
successfully!")
```

12.1 Importing libraries

TensorFlow provides the dataset loader and neural network implementation. NumPy is used to add a batch dimension to a single image and to select the index of the largest output score. Matplotlib is used to display images and graphs.

12.2 Loading the dataset

The load_data() function returns four arrays: training images, training labels, test images, and test labels. The shape of the image arrays confirms that the training set contains 60,000 images and the test set contains 10,000 images, each measuring 28 × 28 pixels.

12.3Normalization

astype('float32') changes the array data type to floating point. Dividing by 255.0 scales the pixel values. The same transformation is applied to training and test images so the model receives inputs in a consistent range.

12.4Modeldefinition

Flatten changes each image into a one-dimensional vector. Dense(128, activation='relu') learns internal representations. Dropout(0.2) randomly sets 20% of the relevant activations to zero during training. Dense(10) generates ten class scores.

12.5Trainingandprediction

model.fit() performs training for five epochs and reserves 10% of the training data for validation. model.evaluate() measures performance on the test set. model.predict() calculates output scores for an input image, and np.argmax() returns the index of the largest score as the predicted digit.

13. TESTING AND EVALUATION

Testing checks whether the program operates as expected and whether the trained model can classify images. The following test cases should be performed after executing the notebook.

Test case	Expected behavior	Actual result
Library import	Libraries load without an error	Successful
Dataset loading	60,000 training and 10,000 test images are loaded	Successful
Image display	A grayscale digit image is displayed	Successful
Preprocessing	Pixel values are converted to 0–1 range	Successful
Training	Training log shows loss and accuracy per epoch	Successful
Evaluation	Test loss and accuracy are printed	Successful, if the evaluation ran
Prediction	Predicted digit is displayed and printed	Digit 7 predicted correctly
Model saving	A .keras model file is created	Confusion matrix generated

13.1Accuracy

Accuracy is calculated as the number of correct predictions divided by the total number of predictions. It is usually reported as a percentage. For example, if a model correctly classifies 9,700 of 10,000 test images, its accuracy is 97%. This is an illustration only; record the value from your own run.

13.2 Loss

Loss represents the model's prediction error according to the selected loss function. Lower loss generally indicates predictions that better match the correct labels, although loss should be interpreted alongside accuracy and other evidence.

13.3 Validation metrics

Validationaccuracy is measured on the 10% portion held out from the training data by validation_split. It helps monitor how well the model performs on examples not used for parameter updates. The separate test set is used for final evaluation.

14. RESULTS AND ANALYSIS

This section must be finalized after the notebook has been executed. Do not report example numbers as if they were obtained from your run. Replace the placeholders below with actual values and paste screenshots from Google Colab.

14.1 Dataset output

1.Observed training image shape: (60000, 28, 28)

2.Observed test image shape: (10000, 28, 28)

3.Screenshot:

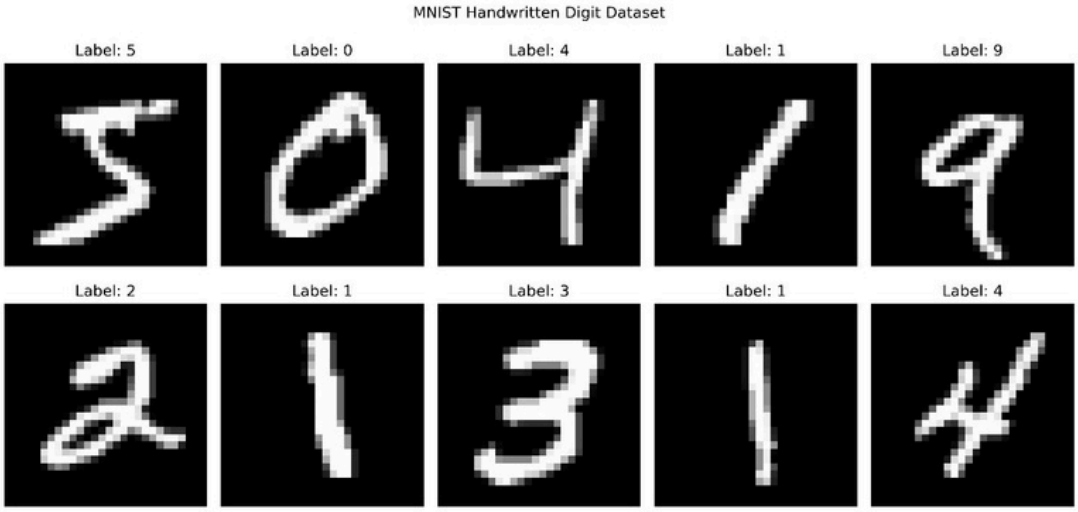


Figure 14.1: Dataset Output

14.2 Training results

Metri c	Actual value from your run
Numberof epochs	5
Final training accuracy	Approx 99.03%
Final validation accuracy	Approx 98.72%
Test accuracy	98.83%
Test loss	0.0357%

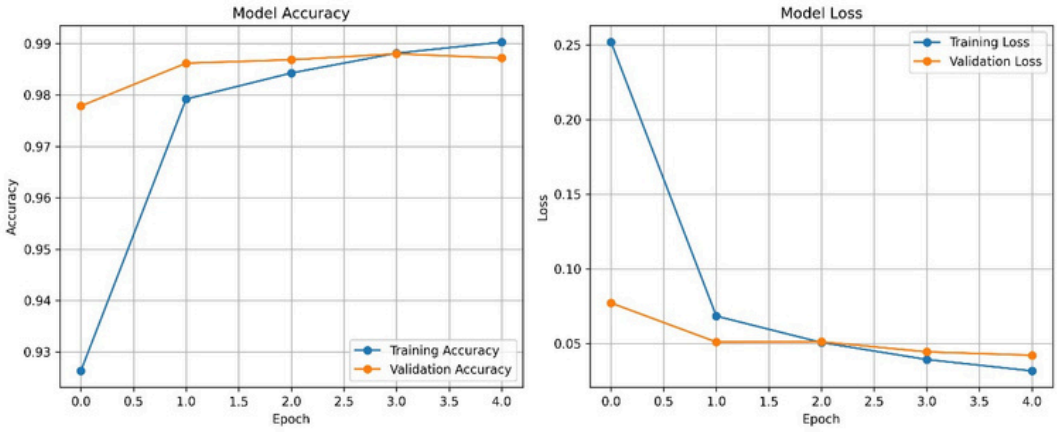


Figure 14.2 Training and Validation Accuracy and Loss

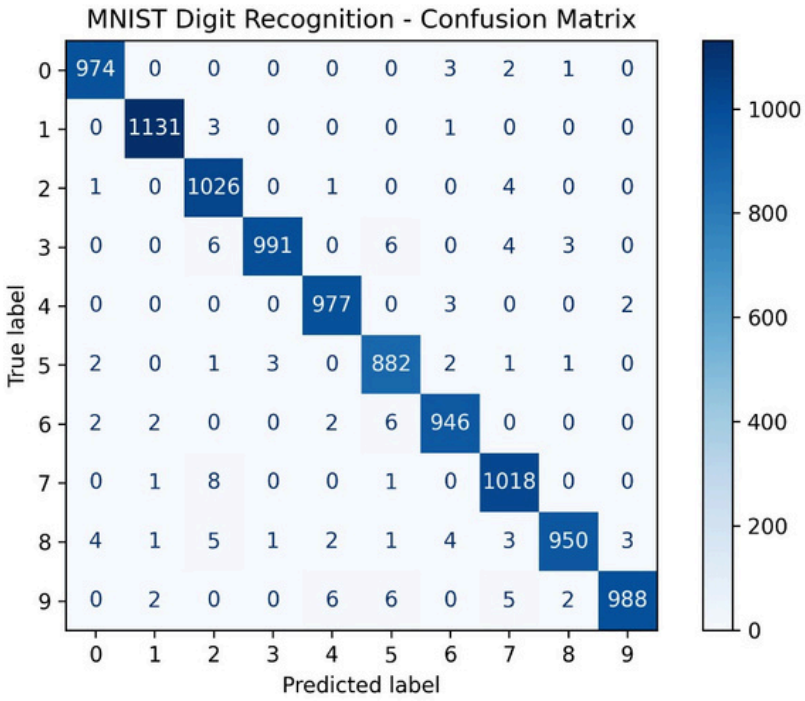


Figure 14.3: Confusion Matrix for Digit Classification

The model was trained for five epochs. The training and validation accuracy increased during training, while the loss decreased. The confusion matrix shows that the model correctly classified 9,883 out of 10,000 test images, achieving a test accuracy of 98.83%.

14.3 Prediction results

Selected test image index: 0

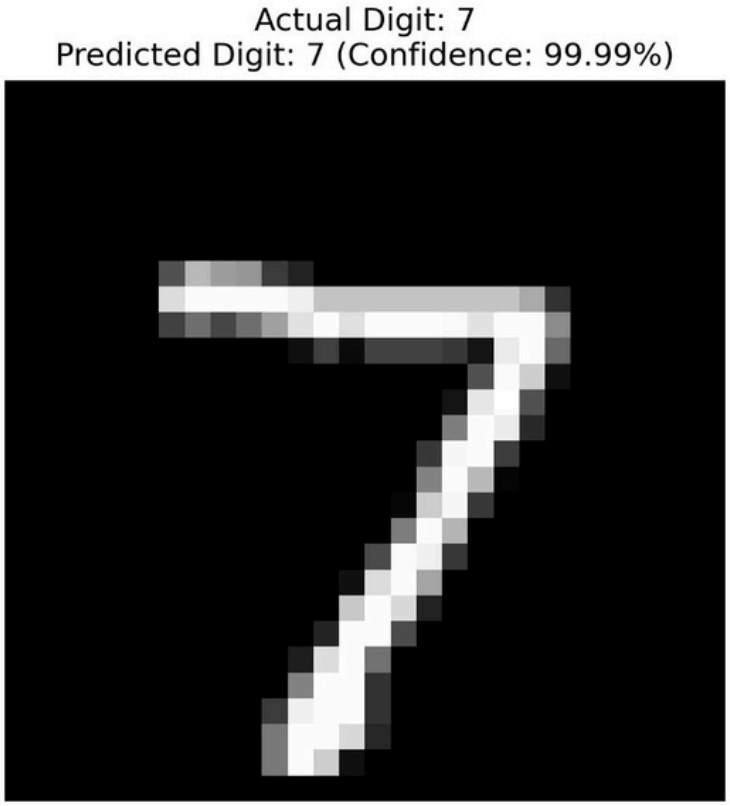
Actual digit label: 7

Predicted digit: 7

Was the prediction correct? YES

Screenshot:

Figure 14.4: Correct Prediction of Handwritten Digit 7.



14.4 Analysis

1. The model achieved a test accuracy of 98.83%, correctly classifying 9,883 out of 10,000 test images.
2. Training and validation graphs show the model's learning progress over five epochs.
3. The selected test image was correctly predicted as digit 7, with 99.99% confidence.
4. The confusion matrix shows that a small number of digits were misclassified.
5. Overall, the model performed well in recognizing handwritten digits from the MNIST dataset.

15. APPLICATIONS

Handwritten digit recognition is useful as a component of larger document-processing systems. This project demonstrates the core classification task, although real applications usually require additional image processing and reliability checks.

- Bank cheque processing: recognizing handwritten numerical fields as part of an automated workflow.
- Postal code recognition: helping read numeric address codes on mail.
- Optical Character Recognition (OCR): identifying characters in scanned documents.
- Form processing: extracting handwritten numbers from application or survey forms.
- Educational tools: demonstrating image classification and giving students examples of AI predictions.
- Historical document digitization: supporting the transcription of handwritten numerical records.
- Data entry support: reducing repetitive manual reading of isolated numerical characters.

15.1Real-worldconsiderations

A real system would need to locate digits within a document, crop and resize them correctly, handle low-quality scans, protect sensitive information, and provide a way to review uncertain predictions. It should be tested on representative real-world data before being used in important decisions.

16. ADVANTAGES AND LIMITATIONS

16.1 Advantages

- Beginner-friendly implementation with a widely used public dataset.
- Demonstrates the complete machine-learning workflow.
- Requires no custom data collection for the basic version.
- Can process many test images quickly after training.
- Provides measurable evaluation through accuracy and loss.
- Can be extended to convolutional neural networks or a simple user interface.

16.2 Limitations

- Themodel istrained on small, centered grayscale images and may not handle real-world photographs reliably.
- The basic model classifies one digit at a time rather than complete words or documents.
- The model may confuse digits with similar shapes.
- The reported accuracy depends on the architecture, training process, and actual run.
- A high test accuracy does not guarantee that every prediction is correct.
- The project does not include a production interface, deployment, or comprehensive error analysis.

16.3 Ethical and responsible use

Automated predictions should not be treated as infallible. If used in a real workflow, uncertain or important predictions should be checked by a human. The model should be evaluated on the intended population and image conditions before deployment.

17. CHALLENGES AND SOLUTIONS

Possible challenge	Reason	Suggested solution
Dataset downloadfails	Internetor temporary connection problem	Check connectionand rerun the dataset-loadingcell.
TensorFlow import error	Environment issue in a local setup	Use Google Colaborinstall a compatible TensorFlow version.
Training takes a long time	Shared compute resources or CPU execution	Wait for completion; use a Colab accelerator if available.
Low accuracy	Model not trained fully or code/data issue	Check normalization, labels, and training log; try more epochs or a CNN.
Prediction differs from actual label	The model made a classification error	Show both values honestly and inspect more test examples.
Notebook variables missing	Runtime restarted or cells ran out of order	Run all cells from the beginning.
Saved model not found	File saved in temporary runtime storage	Download it or save it to Google Drive.

17.1Debuggingapproach

When an error occurs, read the last line of the error message first, then inspect the named code line. Run the smallest relevant cell again after correcting the issue. Keep the original notebook intact until the corrected version works, and save a final copy after successful execution.

18. FUTURE SCOPE

The current implementation is a simple baseline. Several extensions can improve accuracy, usability, and the educational value of the project.

- Convolutional Neural Network (CNN): use convolution and pooling layers to learn spatial image features.
- Custom image input: allow a user to upload an image of a handwritten digit and preprocess it to the required 28 × 28 format.
- User interface: build a small web application using a beginner-friendly framework such as Streamlit.
- Confusion matrix: identify which digit classes are most frequently confused.
- More detailed evaluation: calculate precision, recall, and per-class accuracy where appropriate.
- Model comparison: compare a dense neural network with a CNN using the same train/test split.
- Data augmentation: apply small rotations or shifts during training to improve robustness.
- Deployment: package the trained model for demonstration, with clear limitations and human review where needed.

18.1Proposedimprovement:CNN

A CNN uses filters to detect local patterns such as edges and strokes. These learned features are combined across layers to support classification. CNNs are commonly effective for image tasks and could be compared with the dense network in a future version.

19. CONCLUSION

The Handwritten Digit Recognition project demonstrates how artificial intelligence and neural networks can be used to classify images. The MNIST dataset provides labeled grayscale images for ten digit categories, allowing the complete machine-learning workflow to be implemented without collecting a custom dataset.

The project uses Python, TensorFlow, Keras, NumPy, and Matplotlib. The workflow includes loading the data, exploring a sample image, normalizing pixel values, building a neural network, training the model, evaluating it on test images, plotting accuracy, and predicting a digit.

The project provides practical experience with supervised learning, neural network layers, training and validation, model evaluation, and prediction. Its main limitation is that it works with the specific type of images represented by MNIST and may not directly generalize to all real world handwriting.

After executing the notebook, the student should add the actual accuracy and loss values, insert screenshots of the training process and prediction, and explain the observed results. The project can be improved in the future using convolutional neural networks, custom image uploads, and a simple user interface.

20. REFERENCES

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